

SEMESTER I

BCA- DC101 | Mathematics- I (Calculus & Fourier Transforms)

Course Name: Mathematics - I	
Semester 1	
Course Code	BCA - DC101
Course Title	Mathematics -I (Calculus & Fourier Transforms)
Number of Credits	Total: (L: 3 T: 1)
Total Number of teaching hours	40 Hours
Pre-requisites	Basic Mathematical Concepts
Course Category	Core

Rationale

The accelerated expansion of computing technologies and applications into all our lives means students need to understand the principles of computer science now, more than at any other time. It is necessary for all students to understand the ethical and social role of computer applications in society.

Learning Outcomes

1. To learn basic mathematical concepts.
2. To learn the transform for the use of Image Processing.
3. To learn the basics of transformation Techniques.

Syllabus

Unit	Topics Covered	Hou rs
Unit 1	Differential Calculus	10
1.1	Functions, Elementary Functions, Domain, range, and algebra of Functions, Composite, even and odd Functions	
1.2	Limit, continuity and differentiability: Left and right-hand limit of a function, Algebra of Limits, Indeterminate Forms, L'Hospital Rule, Continuous Function, Properties of Continuous function, Continuity at a point, Continuity in an interval, Derivative of a function by the first principle, Differentiability of a function	

1.3	Differentiation: Differentiation of elementary functions algebraic, trigonometric, inverse trigonometric, logarithmic and exponential functions. Sum, product and quotient functions.	
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1.4	Applications of Derivatives: Rate of change of quantities, applications of derivative in finding errors	
Unit 2	Integral Calculus	10
2.1	Basic Concept, definition, Properties of Integration, Integration as antiderivatives, Integration of Standard Functions, Methods of Integration By substitution, by parts, by partial functions, Definite Integral, some properties of definite integral	
2.2	Application of Integrals: Area under simple curve, Area of a region bounded by a curve and a line, Area between two curves	
Unit 3	Laplace Transform	10
3.1	Laplace transform of elementary functions, Shifting theorems, Transform of derivatives, Differentiation and Integration of transforms, Heaviside unit step and Dirac Delta functions	
3.2	Convolution theorem, Solution of ordinary linear differential equations used in Mechanics, Electric circuits and Bending of beams.	
Unit 4	Fourier Transform and Z Transform	10
4.1	Definition of Fourier transform, Fourier sine and cosine transforms, Fourier integral formula, Parseval's identity, Applications of Fourier transform in solving heat equations.	
4.2	Z transform: Definition, Linearity property, Z transform of elementary functions, Shifting theorems, Initial and final value theorem, Convolution theorem, Inversion of Z transforms, Solution of difference equations by Z transforms.	

References/suggested learning resources

1. Kreyszig E., Advanced Engineering Mathematics, John Wiley, New York, 1999
2. Gerald, C.F., Wheatley P.O., Applied Numerical Analysis, Pearson, 2007.
3. Grewal B.S., Higher Engineering Mathematics, Khanna, New Delhi, 2000.
4. Jain R. K., Iyenger S.R.K., Advanced Engineering Mathematics, Narosa, 2002.
5. Jain R. K., Iyenger S.R.K., Jain M.K., Numerical Methods for Scientific and Engineering Computation, New Age International Publishers, 2012.

BCA-DC102 | Computer Organization and Architecture

Course Name: Computer Organization and Architecture	
Semester 1	
Course Code	BCA-DC102
Course Title	Computer Organization & Architecture
Number of Credits	Total: (L: 2 T: 1 P:2)
Total Number of teaching hours	40 Hours
Pre-requisites	Microprocessors and Digital Electronics
Course Category	Core

Rationale

The accelerated expansion of computing technologies and applications into all our lives means students need to understand the principles of computer science now, more than at any other time. It is necessary for all students to understand the ethical and social role of computer applications in society.

Learning Outcomes

1. To learn the basic structure and operations of a computer.
2. To learn the arithmetic and logic unit and implementation of fixed-point and floating point arithmetic units.
3. To learn the basics of pipelined execution.
4. To understand parallelism and multi-core processors.
5. To understand the memory hierarchies, cache memories and virtual memories.
6. To learn the different ways of communication with I/O devices.

Syllabus

Unit	Title	Hours
Unit 1	Register Transfer Language and Computer Design	10
1.1	Register Transfer and Micro-operations: Register Transfer Language, Register Transfer, Bus and Memory Transfers, Arithmetic Micro-operations, Logic Micro-operations, Shift Microoperations, Arithmetic logic shift unit	6

1.2	Basic Computer Organizations and Design: Instruction Codes, Computer Registers, Computer Instructions, Timing and Control, Instruction Cycle, Memory-Reference Instructions, Register reference instructions, Input - Output Instructions, Design of Accumulator Logic	4
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UNIT 2	Design of CPU and Microprogrammed Control Unit	10
2.1	Microprogrammed Control Unit: Control memory, address sequencing.	3
2.2	Central Processing Unit: Introduction, General Register Organization, Stack Organization, Instruction Formats, Addressing Modes. Difference between RISC and CISC.	3
2.3	Pipeline and Vector Processing: Arithmetic and Instruction pipeline, Vector operations, Matrix Multiplication, memory interleaving.	4
UNIT 3	I/O Organization, Computer Arithmetic and Data transfer	10
3.1	Input-Output Organization: Peripheral Devices, Input-Output Interfaces	4
3.2	Computer Arithmetic: Introduction, Multiplication Algorithms, Division Algorithms for fixed point-members.	3
3.3	Input-Output Organization: Peripheral devices, input-output interface, asynchronous data transfer, modes of data transfer, priority interrupt, direct memory access, input-output processor.	3
UNIT 4	Memory Organization	10
4.1	Memory Organization: Memory Hierarchy, Main Memory, Auxiliary Memory, Associative Memory, Cache Memory, Memory Management Hardware	5
4.2	Multiprocessors: Characteristics of multiprocessor, Interconnection Structure, Interprocessor Communication & Synchronization	5

List of Practical

1. Design and realize all basic gates and universal gates.
2. Design and realize all basic gates from NOR and NAND gates.
3. Realize Half Adder, Full Adder, Half subtractor, Full subtractor
4. Realize a BCD adder
5. Realize a Serial Adder
6. Realize a four bit ALU
7. Realize Master-Slave J K Flip-Flop, using NAND/NOR gates
8. Realize Universal Shift Register
9. Realize Self-Starting, Self Correcting Ring Counter
10. Realize Multiplexer and Demultiplexer
11. Realize Carry Look ahead Adder / Priority Encoder
12. Simulation of PAL and PLA

Learning Approach

To understand the basics of Computer Architecture and Organization and their implementations and applications.

References/suggested learning resources

a) Books

1. Morris Mano, Computer System Architecture, 3rd Edition, Prentice-Hall of India Private Limited, 1999.
2. William Stallings, Computer Organization and Architecture, 4th Edition, Prentice Hall of India Private Limited, 2001
3. Subrata Ghosal, "Computer Architecture and Organization", Pearson 2011
4. Malvino, "Digital Computer Electronics: An Introduction to Microcomputers", McGraw Hill

b) Open source software and website address:

1. Electronics Workbench
2. MultiSim

BCA-DC103 | Problem Solving & Programming in C

Course Name: Problem Solving & Programming in C	
Semester 1	
Course Code	BCA-DC103
Course Title	Problem Solving & Programming in C
Number of Credits	Total: (L: 2 T:1 P: 2)
Total Number of teaching hours	40 Hours
Pre-requisites	NIL
Course Category	Core

Rationale

The accelerated expansion of computing technologies and applications into all our lives means students need to understand the principles of computer science now, more than at any other time. It is necessary for all students to understand the ethical and social role of computer applications in society.

Learning Outcomes

After studying this course, students will be able to:

1. Design a solution to the problem using an Algorithm, Flowchart, and Pseudocode.
2. Use operators and programming constructs in C.
3. Design & Develop functions for the problems.
4. Apply the concept of storage class, arrays, strings, and pointers including dynamic memory allocation, pre-processor directives, structures, union, enumeration, and command-line arguments.
5. Execute the file handling operations in C.

Syllabus

Unit	Title	Hou rs
Unit 1	Problem Solving and Program Design	10
1.1	Problem Solving and Introduction to Programming: Activities involved in Problem Solving, Programming Language, Programming Paradigms, Characteristics of a good program, Classification of Programming Languages, Common programming errors.	
1.2	Program Design: Algorithm, Flowchart, Pseudocode: Introduction, Steps involved and detailed construction, Problems on selection, looping, flow chart analysis.	7
UNIT 2	C Basics, Decision Making and Looping	10
2.1	C basics: Steps involved in execution of a program, C character set, Identifiers and keywords, Data types, constants, variables and arrays, declarations, expressions statements, symbolic constants, compound statements, arithmetic operators, unary operators, relational and logical operators, assignment operators, conditional operators, bit operators, Arithmetic Expressions and its evaluation, Operator precedence and Associativity, Managing Input and Output, Formatted Input and Output.	
2.2	C constructs and flow charts: If statement, if...else statement, Nesting of if...else statement, The else if Ladder, while statement, do....while statement, for statement, switch statement, nested control statement, break operator, continue operator, comma operator, goto statement, flowchart for each construct.	
UNIT 3	C Functions, Storage Classes, Arrays and String Handling	10

3.1	C Functions: Built-in and User Defined Functions, Elements of User Defined Functions, parameter passing mechanisms and returning values, scope, recursion, nesting of functions.	
3.2	Arrays: One-Dimensional, Two-Dimensional and introduction to Multi-Dimensional Array, flow chart, and problem solving on array manipulation.	
3.3	Storage Classes: Automatic, external (global), static & registers.	
3.4	String Handling and library functions: String manipulation functions and other standard library functions	
UNIT 4	Pointers, Pre-processor Directives, Structures and Basic File Management	10
4.1	Pointers: Introduction, Chain of pointers, array & pointer relationship, pointer arithmetic, dynamic memory allocation, pointer to arrays, array of pointers, function returning pointers, pointers to functions, array of pointers to functions.	
4.2	Pre-processor: #include, #define, macros with arguments, the operators # and ##, conditional compilations	
4.3	Structures: Introduction to Structures, Unions and Enumerations, passing structure to functions, Pointer to Structures, bit fields	
4.4	Introduction to File handling [text (ASCII), binary]: Opening, closing, input/output, random access to files. Usage of command-line arguments.	

List of Practical

1. Programs on reading and displaying operands of different types, formatted input and output.
2. Programs on ASCII value of characters.
3. Programs on performing C operations on operands of different data types.
4. Programs on evaluation of arithmetic expressions and operator precedence.
5. Programs on decision making statements and branching: if, if..else, nesting of if..else, The else if Ladder, conditional operator, switch statement, goto, break, continue.

6. Create a user defined function, parameter passing & returning values.
7. Programs on recursion e.g. fibonacci series, factorial etc.
8. Programs on one-dimensional arrays and operations like sum, average etc.
9. Programs on matrices and operations on matrices.
10. Programs on multi-dimensional arrays.
11. Programs on storage classes.
12. Programs on string manipulation.
13. Program to print address of variable & its value.
14. Programs on pointer declaration, initialization & accessing the value pointed to by a pointer, use of pointers in arithmetic expressions.
15. Programs to perform array (single-dimensional & multi-dimensional - up to three dimensions) operations using pointers, array of pointers, function returning pointers & pointers to functions.
16. Programs on dynamic memory allocation.
17. Programs on preprocessor directives.
18. Programs on structures, unions, and enumerations, passing structures to functions, pointer to structures.
19. Programs on File Management.

Learning Approach

To understand the basics of procedure-oriented programming and its applications.

References/suggested learning resources

a) Books

1. Programming in ANSI C, E.Balagurusamy
2. Computer Basics and C Programming, V. Rajaraman
3. The Art of Programming Through Flowcharts & Algorithms, Anil Bikas Chaudhuri.

b) Open source software and website address:

1. Code::Blocks: <https://www.codeblocks.org/>
2. Visual Studio Code: <https://code.visualstudio.com/>
3. Flowgorithm: <http://www.flowgorithm.org/>

Course Name: Information Technologies	
Semester 1	
Course Code	BCA-DC104
Course Title	Information Technologies
Number of Credits	Total: (L:2 T:1 P: 2)
Total Number of teaching hours	40 Hours
Prerequisites	Basic Knowledge of Computer
Course Category	Core

Rationale

The accelerated expansion of computing technologies and applications into all our lives means students need to understand the principles of computer science now, more than at any other time. It is necessary for all students to understand the ethical and social role of computer applications in society.

Learning Outcomes

After studying this course, students will be able to:

1. Describe the major components of information technology applications:
2. Understand the basics of hardware, computer networks, software, data, processes, and people.
3. Describe the different components of a computer network.
4. Demonstrate an understanding of different types of networks.
5. Define “Software Engineering” & Software development Procedure .
6. Discuss the role of databases in IT applications.

Syllabus

Unit	Title	Hours
Unit 1	Introduction to Digital Electronics & System Processing	10
1 . 1	Introduction to Digital Electronics: Digital Signals and Logic gates, Number systems: Binary, octal and hexadecimal number systems, signed binary number, binary arithmetic	
1 . 2	Information System Processing: Information system and information processing, Introduction to Database Management System and its types.	
UNIT 2	Introduction of Information Security and Computer Graphics	10
2 . 1	Introduction to Information Security, Security Goals, Attacks, Security Services and Mechanisms, Cloud Computing.	
2 . 2	Introduction of Computer Graphics: Display Devices: Refresh Cathode Ray Tube, Raster Scan Display, colour display techniques, interactive input/output devices. Animation Applications, Digital Image Processing, Multimedia and Applications.	
Unit 3	Introduction to Operating System and Office Suite	10
3.1	Operating System: Introduction to Operating system, Different types of operating systems and their working, DOS/ Linux commands, File Structure and Storage, Introduction to process management, Windows & Linux Operating System.	
3.2	Office Suite: Introduction, Microsoft Word, Microsoft Excel, Microsoft Powerpoint, OpenOffice	
Unit 4	Introduction to Computer Network	10

4 . 1	Computer Network: Basic Elements of a Communication System, Data transmission media, Digital and Analog Transmission, Network topologies, Network Types (LAN, WAN, and MAN), Introduction to Communication protocols, OSI Model, Internet, IOT.
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List of Practical

1. Program Based on Ms-Word.
2. Program Based on Ms-Excel.
3. Program Based on Ms-Powerpoint.
4. Developing Flow Chart.
5. Basic OS Command.
6. Basic Networking.

Learning Approach

To understand the basics of Computer and Information Technology and their applications

References/suggested learning resources

1. Alex Leon and Mathews Leon, "Fundamentals of Information Technology", Leon Techworld, 2007.
2. P. K. Sinha and Priti Sinha, "Computer Fundamentals", BPB Publications, 2007.
3. Malvino and Leach, "Digital Principles and Application", TMH, 1999.
4. Alex Leon and Mathews Leon, "Introduction to Computers", Vikas Publishing House, 2007.
5. Norton Peter, "Introduction to computers", TMH, 4th Ed., 2006.
6. Simon Haykins, "Communication System", John Wiley & Sons, 2006.
7. B. Basaraj, "Digital Fundamentals", Vikas Publications, 1999.
8. V. Rajaraman, "Introduction to Information Technology", PHI, 2006.

BCA-DC105 | Software Engineering

Course Name: Software Engineering	
Semester 1	
Course Code	BCA-DC105
Course Title	Software Engineering
Number of Credits	Total: (L: 2 T:1 P: 2)
Total Number of teaching hours	40 Hours
Prerequisites	Basic Knowledge of Computer & Software
Course Category	Core

Rationale

The accelerated expansion of computing technologies and applications into all our lives means students need to understand the principles of computer science now, more than at any other time. It is necessary for all students to understand the ethical and social role of computer applications in society.

Learning Outcomes

After studying this course, students will be able to:

1. To understand the iterative implementation of software projects.
2. To analyze projects using use case modelling tools.
3. To develop solutions for real-life cases using design models and patterns.
4. To understand and implement project design requirements for user interface, data layer, and system controls.
5. To apply modern case tools to develop solutions.

Syllabus

Unit	Title	Hours
Unit 1	Introduction, Process Model, Requirement Analysis & Specification	10
1.1	Introduction: Software Crisis, Software Processes & Characteristics, Software life cycle models, Waterfall, Prototype, Evolutionary and Spiral Models	
1.2	Software Requirements analysis & specifications: Requirement engineering, requirement elicitation techniques like FAST, QFD, Requirements analysis using DFD(with case studies), Data dictionaries & ER Diagrams, Requirements documentation, Nature of SRS, Characteristics & organization of SRS.	
Unit 2	Software Project Management	10
2.1	Software Project Management Concepts: The Management spectrum, Key Objectives of Effective Management	
2.2	Software Project Planning: Size Estimation like lines of Code & Function Count, Cost Estimation Models, COCOMO, Risk Management.	
Unit 3	Software Design & Metrics	10
3.1	Software Design: Cohesion & Coupling, Classification of Cohesiveness & Coupling, Layered arrangement of modules, Function Oriented Design, Object Oriented Design	
3.2	Software Metrics: Software measurements: What & Why, Token Count, Halstead Software Science Measures, Design Metrics, Data Structure Metrics.	

Unit 4	Software Testing, Reliability & Maintenance	10
4.1	Software Testing: Code Review, Testing Process, Types of Testing, Functional Testing, Structural Testing, Test Activities, Unit Testing, Integration Testing and System Testing(Performance Testing and Error Seeding), Debugging Activities.	
4.2	Software Reliability: Failure and Faults, Reliability Models: Basic Model software Quality Standards SEI & CMM Software Maintenance: Management of Maintenance, Maintenance Process, Reverse Engineering, Software Re-engineering, Configuration Management, Documentation	

List of Practical

1. Prepare Problem Statement
2. Prepare Feasibility Report
3. Prepare DFD, ERD
4. Prepare Use Case Diagram
5. Prepare Design Document
6. Prepare Structure Chart, Decision Table
7. Prepare Test Case & Test Suit
8. Prepare Test Report

Learning Approach

To understand the basics of software engineering and its applications.

References/suggested learning resources

1. K. K. Aggarwal & Yogesh Singh, "Software Engineering", 2nd Ed., New Age International, 2005.
2. Rajib Mall, "Fundamental of Software Engineering", 3rd Edition, PHI Learning Private Limited
3. I. Sommerville, "Software Engineering", 9th Edition, Pearson Edu.
4. Jalote Pankaj , "Software Engineering", Narosa Publication
5. R. S. Pressman, "Software Engineering – A practitioner's approach", 5th Ed., McGraw Hill Int. Ed., 2001.
6. James Peter, W. Pedrycz, "Software Engineering: An Engineering Approach", John Wiley & Sons.